

PRACTICAL 8

Perform the data clustering using clustering algorithm.

k-means clustering using R

#apply K means to iris and store result

newiris <- iris

newiris\$Species <- NULL

(kc <- kmeans(newiris,3))

K-means clustering with 3 clusters of sizes 21, 96, 33

Cluster means:

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
1	4.738095	2.904762	1.790476	0.3523810
2	6.314583	2.895833	4.973958	1.7031250
3	5.175758	3.624242	1.472727	0.2727273

Clustering vector:

```
[1] 3 1 1 1 3 3 3 3 1 1 3 3 1 1 3 3 3 3 3 3 3 3 1 1 3 3 3 1 1 3 3 3 1 3 3 3 1
[40] 3 3 1 1 3 3 1 3 1 3 3 2 2 2 2 2 2 1 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[79] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 1 2 2 2 2 1 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
[118] 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2 2
```

Within cluster sum of squares by cluster:

```
[1] 17.669524 118.651875 6.432121
(between_SS / total_SS = 79.0 %)
```

Available components:

[1]	"cluster"	"centers"	"totss"	"withinss"	"tot.withinss"
[6]	"betweenss"	"size"	"iter"	"ifault"	

#Compare the Species label with the clustering result

table(iris\$Species,kc\$cluster)

	1	2	3
setosa	17	0	33
versicolor	4	46	0
virginica	0	50	0

#Plot the clusters and their centers

plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc\$cluster)

points(kc\$centers[,c("Sepal.Length","Sepal.Width")],col=1:3,pch=8,cex=2)

dev.off()

#Plot the clusters and their centre

plot(newiris[c("Sepal.Length","Sepal.Width")],col=kc\$cluster)

